

EXPRESSION OF INTEREST (EOI)

**Title of Consulting Service: Generation of High-Resolution DEM and Ortho photo Map with application of
Airborne LiDAR Surveying and Mapping**

Method of Consulting Service: International

**Project Name : Generation of High-Resolution DEM and Ortho photo Map with
application of
Airborne LiDAR Surveying and Mapping**

EOI : SD/ICS/QCBS/01/2081/082

Office Name: Survey Department

Office Address: Minbhawan Kathmandu Kathmandu

Funding agency : Government Budget

Abbreviations

CV	-	Curriculum Vitae
DO	-	Development Partner
EA	-	Executive Agency
EOI	-	Expression of Interest
GIID	-	Geographic Information Infrastructure Division
GON	-	Government of Nepal
PAN	-	Permanent Account Number
PPA	-	Public Procurement Act
PPR	-	Public Procurement Regulation
TOR	-	Terms of Reference
VAT	-	Value Added Tax

A handwritten signature in black ink, appearing to be a stylized name, possibly "G. G. G.", with a horizontal line extending from the right side.

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A. Request for Expression of Interest

Request for Expression of Interest

Government of Nepal (GoN)

Name of Employer: Survey Department

Date: 23-04-2025 00:00

Name of Project: Generation of High-Resolution DEM and Ortho photo Map with application of Airborne LiDAR Surveying and Mapping

1. Government of Nepal (GoN) has allocated fund toward the cost of Generation of High-Resolution DEM and Ortho photo Map with application of Airborne LiDAR Surveying and Mapping and intend to apply portion of this fund to eligible payments under the Contract for which this Expression of Interest is invited for International consulting service
2. The Survey Department now invites Expression of Interest (EOI) from eligible consulting firms (“consultant”) to provide the following consulting services: Generation of High-Resolution DEM and Ortho photo Map with application of Airborne LiDAR Surveying and Mapping
3. Interested eligible consultants may obtain further information and EOI document free of cost at the address Survey Department, Survey Department
Minbhawan
Kathmandu, Kathmandu
Bagmati Province
Nepal during office hours on or before 08-05-2025 12:00 or visit e-GP system www.bolpatra.gov.np/egp or visit the client’s website dos.gov.np
4. Consultants may associate with other consultants to enhance their qualifications.
5. Expressions of interest shall be delivered online through e-GP system www.bolpatra.gov.np/egp on or before 08-05-2025 12:00
6. In case the last date of obtaining and submission of the EOI documents happens to be a holiday, the next working day will be deemed as the due date but the time will be the same as stipulated.
7. EOI will be assessed based on Qualification 40.0 %, Experience 45.0 %, and Capacity 15.0 % of consulting firm and key personnel. Based on evaluation of EOI, only shortlisted firms will be invited to submit technical and financial proposal through a request for proposal.
8. Minimum score to pass the EOI is 70

B. Instructions for Submission of Expression of Interest

Instructions for Submission of Expression of Interest

1. Expression of Interest may be submitted by a sole firm or a joint venture of consulting firms and the maximum number of partners in JV shall be limited to three.
2. Interested consultants must provide information indicating that they are qualified to perform the services (descriptions, organization and employee and of the firm or company, description of assignments of similar nature completed in the last 7 years and their location, experience in similar conditions, general qualifications and the key personnel to be involved in the proposed assignment).
3. This expression of interest is open to all eligible consulting firm/person/ company/ organization.
4. In case, the applicant is individual consultant, details of similar assignment experience, their location in the previous 4 years and audited balance sheet and bio data shall be considered for evaluation.
5. The assignment has been scheduled for a period of 21 Months. Expected date of commencement of the assignment is 07-07-2025.
6. A Consultant will be selected in accordance with the QCBS method.
7. Expression of Interest should contain following information:
 - (i) A covering letter addressed to the representative of the client on the official letter head of company duly signed by authorized signatory.
 - (ii) Applicants shall provide the following information in the respective formats given in the EOI document:
 - EOI Form: Letter of Application (Form 1)
 - EOI Form: Applicant's Information (Form 2)
 - EOI Form: Work Experience Details (Form 3(A), 3(B) & 3(C))
 - EOI Form: Capacity Details (Form 4)
 - EOI Form: Key Experts List (form 5).
8. Applicants may submit additional information with their application but shortlisting will be based on the evaluation of information requested and included in the formats provided in the EOI document.
9. The Expression of Interest (EOI) document must be duly completed and submitted by electronically only using the forms and instructions provided by the system.
10. The completed EOI document must be submitted on or before the date and address mentioned in the "Request for Expression of Interest". In case the submission falls on public holiday the submission can be made on the next working day. Any EOI Document received after the closing time for submission of proposals shall not be considered for evaluation.

C. Objective of Consultancy Services or Brief TOR

TERMS OF REFERENCE (TOR)
For
Procurement of International Consulting Services -
Generation of High-Resolution DEM and Ortho
photo Map with application of
Airborne LiDAR Surveying and Mapping
SD/ICS/QCBS/01/2081/2082



Terms of Reference (ToR)

For

Procurement of International Consulting Services - Generation of High Resolution DEM and Ortho photo Map with application of Airborne LiDAR

A. Background

Survey Department, Ministry of Land Management, Cooperatives and Poverty Alleviation, Government of Nepal is the national mapping organization, is mandated to provide digital topographical mapping and other related base map services as well as information. Geographic Information Infrastructure Division (GIID), under Survey Department, was initially set up as Census Mapping Project in 1998 AD, which was later upgraded to National Geographic Information Infrastructure Project (NGIIP) later in 2002 AD. The topographic base maps prepared from the FINIDA and JICA funded topographic mapping projects were also updated and integrated to National Topographic Database (NTDB) under the project. Later upgraded to Geographic Information Infrastructure Division (GIID) in 2015 AD, the division is now one of four divisions under Survey Department. It mainly acts as the National Spatial Information Infrastructure. Since then, it has been contributing as the national data clearinghouse, that offers spatial data and services through National Geoportal. Since its upgrade, one of the major achievement has been successful commencement of Airborne LiDAR survey of 20,000 square kilometers area of Western Terai of Nepal.

LiDAR, abbreviation for Light Detection and Ranging, is a remote sensing method that uses light in the form of a pulsed laser to measure ranges (variable distances) to the Earth. Airborne LiDAR consists of high frequency laser scanner that records the time difference between the emission of the laser pulses and the reception of the reflected signal from the environment. These light pulses combined with other data recorded by the airborne system generate precise, three-dimensional information about the shape of the Earth and its surface characteristics. LiDAR creates highly accurate and detailed models of the earth's surface with horizontal and vertical accuracy down to centimeters level. Another major component of the mission is to acquire aerial photographic images of the region to produce high resolution orthophotograph. These products are the backbone to extraction of details at higher accuracy level for large scale mapping.

LiDAR systems allow mapping professionals to examine both natural and manmade environments with accuracy, precision, and flexibility. LiDAR has been used in variety of platforms and implication including digital elevation models for use in geographic information systems, to assist in emergency response operations, and in many other applications.

B. Objective of the Service

The major objective of the assignment is to prepare

- i. Contour at 0.5m interval,
- ii. Digital Elevation Model (DEM) at 1m resolution,
- iii. Digital Surface Model (DSM) at 1m resolution,
- iv. Orthophoto at resolution (GSD) of 0.10 m, of project area,
As prime outputs and
- v. Control Points Establishment by Differential GNSS
As supplementary output.



C. Specification of LiDAR Surveying and Mapping

i. Coverage

Area of interest (AOI) for the project will be marked in the map (Appendix A) and in digital shape file format will be provided later. Except for Nepalese International Border, area of interest (AOI) should be buffered by 150 meters.

ii. LiDAR Point Cloud Density

LiDAR Point Cloud Density must be at least **10 points/m²** (*except in water bodies*), however considering terrain and other conditions, point density may be increased to maintain desired accuracy for achieving the quality end products whenever required.

iii. Ground sample distance (GSD) of the aerial photographs

The digital aerial photographic images should be captured with a GSD of 0.08 m. or better at nadir.

iv. Photographic Coverage

The forward overlap (forelap) between successive exposures in each run should be minimum 70 percent and the lateral overlap (sidelap) between adjacent strips should be minimum 30 percent. However, these values may be increased to maintain desired accuracy.

v. Fundamental Spatial Accuracy

The vertical accuracy or the **root mean square error (RMSE)** of the end products should be within a range of ± 0.25 m or better and the horizontal accuracy or the **root mean square error (RMSE)** of orthophoto should not exceed three times the pixel size.

vi. Coordinate Reference System (CRS):

- **Vertical Coordinate System**

Primary vertical coordinate system will be ellipsoidal height based on WGS 84. Data that are related to elevation should be adjusted to local height datum using geoid model.

- **Horizontal Coordinate System**

The primary horizontal coordinate system will be UTM projection based on WGS84. The deliverables should also be transformed to National Coordinate System.

vii. Ground Control System

- Maximum distance between the Reference GNSS station on the ground and airborne GNSS units must not exceed 25 kilometers.
- All control survey data used or derived from this assignment must be supplied to ensure independent Quality Assurance (QA) of the survey operations. It is therefore essential that all primary ground stations should be visible in photographs.
- The primary ground control and check point surveys must be referenced to survey control marks with geodetic control points (in terms of coordinates and height) demarcated by survey department.
- Elevation data must be validated and corrected for systematic errors to ensure accuracy specifications are met. Documentation must describe how this has been achieved.



viii. **LiDAR Data acquisition details**

- A Draft Pre-flight Agreement with provision whereby LiDAR data over the Study Area must not be collected during any period, e. g. flood, adverse weather etc., where extent of LiDAR ground returns in any part of the Study Area is likely to be significantly compromised.
- A Draft Pre-Flight Agreement will include provision whereby the Principal is notified of each proposed LIDAR collection flight with sufficient notice to enable consultation between Principal and the contractor to determine if data capture by the Contractor should proceed.
- Collections in high relief terrain are expected to require greater overlap. Any data with gaps between the geometrically usable portions of the swaths will be rejected.
- The spatial distribution of geometrically usable points is expected to be uniform and free from clustering. In order to ensure consistent data densities throughout the project area:

Environmental conditions for data capture:

- Cloud and fog free between the vehicle and the ground.
- Floodplain/wetland data must be captured during times of base flow and outside of significant surface inundation due to natural events and/or regulated environmental flows.
- Details of the vehicle (- helicopter or aeroplane), navigation and mission planning activities for LIDAR and digital photography acquisition. Include details of whether the photography and LIDAR will be acquired during the same mission (i.e. from the same vehicle - helicopter or aeroplane) or from separate missions. If separate missions are required to satisfy the respective resolution requirements for LIDAR and photography (eg. Due to flying height constraints) then specify alternative mission scenarios based on a sensor type and platform.

ix. **Orthophoto**

- RGB and single band NIR orthophoto at resolution of 0.1 m, should be generated by stereoplotting.
- Tiled delivery, as per Data Supply Specifications.

x. **Digital Surface Model (DSM) (orthometric)**

- Digital Surface Model (DSM) at a resolution of 1m grid.
- The DSM should be generated from the “first return” LiDAR mass point data. This will include ground and non-ground points such as vegetation and buildings.
- The DSM generation should employ a Point to TIN and TIN to Raster process with Natural Nearest Neighbour interpolation.
- Void areas (i.e., areas outside the project boundary but within any tiling scheme) shall be coded using a unique “NODATA” value.
- Tiled delivery, as per Data Supply Specifications.



xi. **Digital Terrain Model (DTM)(orthometric)**

- 1m grid bare earth Digital Elevation Model (DEM).
- The DEM should be generated from the LiDAR mass point data classified as “Ground” only, so that it defines the “bare earth” ground surface.
- The DEM generation should employ a Point to TIN and TIN to Raster process with Natural Nearest Neighbour interpolation.
- Void areas (i.e., areas outside the project boundary but within any tiling scheme) shall be coded using a unique “NODATA” value
- Tiled delivery, as per Data Supply Specifications.

xii. **Differential GNSS Data collection**

- GNSS data for all base station occupations in excess of 4 hours is to be provided in RINEX version (Receiver Independent Exchange Format) 2.10 or recent format.
- GNSS observation log sheets which include the following details:
 - a. Survey mark id
 - b. Occupation time & date
 - c. Antenna height measurements
 - d. Instrument /antenna types & serial numbers.
- The GNSS observation log sheets should be provided in pdf format or Excel spreadsheet if data is captured digitally.

xiii. **Metadata**

- For each supplied data product a complete metadata statement consistent with the ISO Standard.
- Metadata must be provided with final deliveries.
- The job will not be signed off by Contract Authority until the metadata is satisfactorily supplied.

xiv. **Spatial Accuracy Validation**

- The fundamental vertical accuracy of the point cloud dataset must be determined with check points located only in open, relatively flat terrain, where there is a very high probability that the sensor will have detected the ground surface.
- The vertical accuracy of the point cloud dataset is to be tested using a TIN surface constructed from bare-earth LiDAR points compared against ground survey check points.
- Check points are to be surveyed independently of any LiDAR GNSS operations.
- The number of check points (locations) is dependent on the extent of the survey. The following strategy should be used as a guide:



- a. Check points must be established to adequately cover the full extent of the survey area, and be representative of the project area landscape.
- b. A minimum of 1 point per 50km² in 7 km * 7 km grid in different vegetation types and distributed evenly over the Project area.
- The proposed check point survey design must be submitted with the quotation, and approved by the Contract Authority prior to implementation. Acceptance of the post-survey spatial accuracy report discussed above is dependent on the quality, number and distribution of these check points.
- In the above circumstances a “compiled to meet” statement of horizontal accuracy at 95 percent confidence should be reported.

xv. **Intermediate Deliverables**

- Deliverable 1: mission flight plans in ESRI shape file format.
- Deliverable 2: Report on planned ground control, check points – including details of connections to national control network - in Word and ESRI Shape files.
- Deliverable 3: Report on planned check points to be collected by the Consultant At least 1 check points must lie within 7 km X 7 km - Word and ESRI Shape files.

xvi. **Final Deliverables**

i) **LiDAR Deliverables**

- Deliverable 1: Raw LiDAR (point cloud) survey data comprising all returns (1st, 2nd, etc..., and last return), representing all ground and non-ground returns and including overlapping swathes - in ASPRS LAS (LAS v 1.2 or stable), file format X, Y, Z coordinates, return intensity value, time and date for each point. All points retained, and classified as per LAS v 1.2 or stable (See: <http://www.lasformat.org/>)
- Trajectory files providing information about the scanner’s position while capturing the point cloud.
- Deliverable 2: Digital Surface Model (DSM) and Digital Terrain Model (DTM) of 1m grid resolution - DEM should be supplied in geoTiff format in original WGS 84 and National Coordinate System.
- Deliverable 3: Contour of 0.5 m interval - Contour should be supplied in ESRI shape file original WGS 84 and National Coordinate System..

ii) **Orthophoto**

- Deliverable 4: Near-natural colour-balanced digital photography for photogrammetry application. Georeferenced, but not ortho-rectified. 0.10 m. ground sample distance (GSD) - Photographs with GPS INV data, Campfile and IMG/TIFF/GEOTIFF format - Red, Green & Blue bands.
- Deliverable 5: Ortho rectified multispectral digital photograph with 0.10 m. GSD. – GEOTIFF Red, Green, Blue.
- Deliverable 6: Colour balanced mosaic with 0.10 m. GSD - Red, Green and Blue bands.



iii) **Documents**

- Flight Data Storage (FDS) report: A report shall be delivered with the raw data (on FDS) giving the following information:
 1. Project name.
 2. Number of images.
 3. FDS metadata report
 4. Year(s), month(s) and day(s) of raw data.
 5. Camera type and Camera Calibration Information.
 6. Run number and flight direction.
 7. Aircraft type and identification.
 8. Names of pilot(s), navigator and photographer.
 9. Start and end time for each run in local time.
 10. Computed altitude above mean sea level (true altitude).
 11. GSD of output images.
 12. Weather conditions – cloud type, degree of haze, turbulence etc.
 13. General comment on quality.
- Output Images Report: A report shall be delivered with the output images giving the following information:
 1. Project name
 2. Number of images
 3. Year(s), month(s) and day(s) of raw data.
 4. Camera type and Camera Calibration Information.
 5. Strip number and flight direction.
 6. Aircraft type and identification.
 7. Names of pilot(s), navigator and photographer.
 8. Start and end time for each run in local time.
 9. Computed altitude above mean sea level (true altitude).
 10. Type and GSD of output images.
 11. Date of processing
 12. Operator of Post Processing
 13. Weather conditions - cloud type, degree of haze, turbulence etc.
 14. General comment on quality
- Other material to be delivered
 1. Plot of planned flight missions
 2. Plot of finished flight missions
 3. Aerial survey report
- Point cloud classification report
 1. Software used
 2. Point cloud classification rules
 3. Detail on manual classification
 4. Point cloud classification accuracy assessment data
- Detail Final report with all necessary details.

xvii. **Project Period:**

Total time duration of project is 21 months.

xviii. **Team Composition of Key Experts**



S.N.	Position	Qualification	Experience	Additional Qualification	Number of Person
1.	Team Leader	At Least, Master's Degree in Geomatics engineering/Geoinformatics or GIS and Remote sensing or Environmental Science/ Environmental Engineering or Forestry or Natural Resource Management or Applied Earth Science urban/rural/regional Planning or Geography with GIS and Remote Sensing, Photogrammetric. Preferable Ph.D.in one of the above mentioned subjects.	At least 5 projects of relevant work experience in LiDAR Mapping and Surveying field. After completion of Basic Degree .	With GIS and Remote Sensing included in course or at least six months mandatory training in GIS and Remote sensing or having post-graduate diploma in GIS or Remote sensing as an extra academic achievement in addition to the specified master degree if GIS and RS are not included in course.	1
2.	Deputy Team Leader	At Least, Master's Degree in Geomatics engineering / Geo-informatics or GIS and Remote sensing or Environmental Science/ Environmental Engineering or forestry or Natural Resource Management or Applied Earth Science urban/rural/regional Planning or Geography with GIS and Remote Sensing, Photogrammetric. Preferable Ph.D.in one of the above mentioned subjects.	At least 4 projects of relevant work experience in related field. After completion of Basic Degree	With GIS and Remote Sensing included in course or at least six months mandatory training in GIS and Remote sensing or having post-graduate diploma in GIS or Remote sensing as an extra academic achievement in addition to the specified master degree if GIS and RS are not included in course.	1
3.	Senior Surveyor/ Geomatic Engineer	Having Completed at least Senior Surveying Course (or equivalent to senior surveying Course)/or B.E. in Geomatics Engineering/Survey engineering.	At least 2 projects of relevant work experience in related field after Completion of the relevant course After completion of Basic Degree	-	2
4.	GIS /RS Expert	At least having BE in geomatic engineering preferable master's Degree with GIS/RS included and studied in course or having PG diploma in GIS/RS after the completion of Bachelor's Degree.	At least 2 projects of relevant work experience in related field after completion of relevant course. After completion of Basic Degree	-	2
5.	GPS Operator	One year Junior Surveyor Course or Diploma in Geomatics.	At least 1 projects of relevant work experience in related field. After completion of Basic Degree	-	4
xix. <u>Non Key Expert</u>					
6.	Field Enumerator	At least intermediate level.		-	4

D. LAB Setup:

To establish a fully equipped lab for LiDAR data processing and analysis, the bidder is required to provide the necessary hardware and software as outlined. The lab setup will include **four high-performance workstations** to handle complex geospatial data efficiently. Additionally, the lab will feature a comprehensive suite of ArcGIS software licenses to support a wide range of geospatial applications, including 3D analysis, data

validation, geostatistical modeling, and advanced mapping capabilities. The bidder must ensure that all software licenses are genuine, correctly installed, and fully compatible with the workstations provided. The table below outlines the specific components required for the lab setup:

Category	Item	Quantity
Workstations	High-Performance Workstation	4
Processing and Mapping Software	ArcGIS Pro Professional Plus (64-bit) – Perpetual License (or equivalent). The software version released should be of year 2024 or latest. The software must include the following extensions:	1
	3D Analyst	1
	Data Reviewer	1
	Geostatistical Analyst	1
	Image Analyst	1
	Locate XT	1
	Network Analyst	1
	Publisher	1
	Spatial Analyst	1
	Workflow Manager	1

Note: The lab is designed to meet the demands of advanced geospatial analysis and mapping projects. The bidder must ensure timely delivery, installation, and proper configuration of all components for the lab to be operational.

Specification of Workstation:

SN	Components	Requirement
1	CPU	AMD Ryzen Threadripper 7960X 24-Core, 48-Thread 5.3GHz Max
2	RAM	128GB ECC RDIMM - Lifetime Warranty
3	Video Card	Nvidia RTX 4060 Ti 16GB GDDR6 4352 CUDA Cores
4	CPU Cooler	Threadripper Water Cooler - 3x120mm Fans - Sealed & Maintenance Free - 1 Yr Warranty - Highest Performance Option
5	Storage	PCIe NVMe SSD(s) High Performance SSD – 4,000GB - Up to 7,250 MBps - 1 Yr Warranty - NVMe PCIe 4.0
6	OS	Factory Enabled Genuine Windows 11 professional 64 bit
7	Warranty	1-Years Parts, 1-Year Advanced Parts Replacement
8	Accessories	Mouse and Keyboard
9	Monitor	27” UHD 4K



E. Training

Training Should be provided as proposed by the survey department. The table provides a clear summary of the objectives, number of participants, and the costs to be covered for both the Course on Lidar Scanning (1 month) and the Exposure Visit (1 week), ensuring bidders have a detailed understanding of the requirements for their proposals.

Training Component	Objective	Number of Participants	Costs to be Covered
Course on Lidar Scanning and Data Processing (1 month)	To equip participants with technical knowledge and practical skills in Lidar technology as specified in the course below.	5	- Course Fee - Round-Trip Travel Costs - Accommodation Rent - Daily Allowance (meals and local transport) - Travel Insurance
Exposure Visit (1 week)	To provide practical insights and exposure to best practices in Lidar application for urban settings.	4	- Round-Trip Travel Costs - Hotel Accommodation - Daily Allowance (meals and local transport) - Travel Insurance

Course Overview:

This comprehensive course, offered by an International Institute or Academia certified in LiDAR data acquisition and processing, must cover the principles of LiDAR technology with a strong emphasis on data acquisition and processing. The course should provide a thorough understanding of LiDAR technology, including its principles, data acquisition techniques, and processing methods. Participants must gain hands-on experience with LiDAR datasets, using Terrasolid software and other industry-standard tools. They should learn how to apply various processing algorithms for generating and analyzing orthophotos and Digital Elevation Models (DEMs), and explore real-world applications in fields such as geography, environmental science, urban planning, and forestry.

Course Objectives:

- Understand the fundamental principles of LiDAR technology.
- Acquire skills in LiDAR data collection and processing.
- Generate and analyze orthophotos and Digital Elevation Models (DEMs).
- Utilize software tools for LiDAR data visualization and analysis.
- Interpret LiDAR data for various applications.

Modules:

Module 1: Introduction to LiDAR

- **Lecture 1: Fundamentals of LiDAR Technology**
 - Cover principles of LiDAR.



- Explain types of LiDAR systems (Terrestrial, Airborne, Mobile).
- **Lecture 2: History and Development of LiDAR**
 - Discuss the evolution of LiDAR technology.
 - Highlight key milestones and advancements.
- **Practical Session:** Provide an overview of LiDAR equipment and software.

Module 2: LiDAR Data Acquisition

- **Lecture 1: Planning LiDAR Surveys**
 - Discuss survey design considerations.
 - Explain flight planning for airborne LiDAR.
- **Lecture 2: Data Collection Methods**
 - Teach terrestrial and mobile LiDAR scanning techniques.
 - Cover airborne LiDAR data collection.
- **Lecture 3: Best Practices in Data Acquisition**
 - Ensure quality assurance and control.
 - Discuss safety and regulatory considerations.
- **Practical Session:** Conduct a LiDAR survey (fieldwork).

Module 3: LiDAR Data Processing

- **Lecture 1: Introduction to LiDAR Data Processing**
 - Provide an overview of processing workflows.
 - Introduce software tools for processing.
- **Lecture 2: Pre-processing of LiDAR Data**
 - Cover data import and format conversion.
 - Teach noise reduction and filtering techniques.
- **Lecture 3: Point Cloud Processing with Terrasolid**
 - Teach classification of LiDAR points.
 - Demonstrate generating Digital Elevation Models (DEMs).
- **Lecture 4: Orthophoto Generation**
 - Explain integration of LiDAR and imagery.
 - Demonstrate creating orthophotos from aerial imagery.
- **Practical Session:** Conduct hands-on LiDAR data processing using Terrasolid software.

Module 4: Applications of LiDAR

- **Lecture 1: Environmental Applications**
 - Teach forest mapping and biomass estimation.
 - Cover flood modeling and watershed analysis.
- **Lecture 2: Urban and Infrastructure Applications**
 - Teach urban planning and 3D city modeling.
 - Cover transportation and infrastructure management.
- **Practical Session:** Provide case studies and project work.

Module 5: Advanced LiDAR Processing Techniques

- **Lecture 1: Advanced Features of Terrasolid**
 - Teach feature extraction and 3D modeling.
 - Cover automation and scripting in Terrasolid.
- **Lecture 2: Machine Learning and LiDAR**
 - Integrate machine learning for LiDAR data analysis.
 - Demonstrate automated feature extraction.
- **Lecture 3: Multi-sensor Data Fusion**
 - Teach combining LiDAR with other geospatial data (e.g., imagery, GPS).
 - Discuss benefits and challenges of data fusion.
- **Practical Session:** Conduct advanced processing exercises.

Note: All expenses for travel, insurance, accommodation, and the course fee for Five participants must be covered by the consultant and ensures participants can fully engage in the course focusing on acquiring valuable skills and knowledge in LiDAR technology.

F. Reporting Stages

Reports should be provided at various stages for regular monitoring of the project and engagement of stakeholders. The Consultants shall submit the following reports in the specified timeframe as mentioned below:

Report Type	Content	Submission Timeline	Copies Required
Inception Report	- Feasibility analysis and survey plan. - Identification of priority areas and logistical considerations.	Within 15 days from the date of contract.	3 hard copies with softcopy.
Interim Progress Reports	- Updates on data collection milestones and preliminary results. - Identification of challenges and corrective measures.	Every quarter.	3 hard copies with softcopy.
Final Report	- Comprehensive dataset delivery (including DEMs, DSMs, and classified point clouds). - Summary of findings, applications, and recommendations for utilization. - Draft report submission required before the final report.	Draft: Before Final Report Final: Upon draft approval.	6 hard copies with softcopy.
Post-Survey Review	- Evaluation of project outcomes and lessons learned. - Suggestions for future updates or maintenance of datasets.	After the completion of the survey.	2 hard copies with softcopy.

G. Payment Schedule



- **First Payment:** Ten (10) percent of the contract price shall be paid after submission of Satisfactory Inception report acceptable to the Client
- **Second Payment:** Fifteen (15) percent of the contract price shall be paid within Fifteen (15) days of submission of Satisfactory DGPS field survey report and Flight Plan Report acceptable to the client.
- **Third Payment:** Ten (10) percent of the contract price shall be paid after supply and installation of high performance workstations and Software.
- **Fourth Payment:** Twenty (20) percent of the contract price shall be paid within Thirty (30) days of submission of Satisfactory LiDAR data capture report and raw data acceptable to the client
- **Fifth Payment:** Twenty (20) percent of the contract price shall be paid within Thirty (30) days of submission of Satisfactory Draft Deliverables on the assignments as per TOR, Technical Specification acceptable to the client.
- **Final Payment:** Twenty Five (25) percent of Contract Price shall be made within thirty (30) days following the Client's written confirmation of acceptance of the Final Deliverables, provided that such deliverables are submitted in full compliance with the Terms of Reference, technical specifications, and all comments and recommendations made during the review of the draft submissions.
- Note:
 - ❖ All Payment shall be made in Nepalese Currency.
 - ❖ No additional Charges, taxes or penalty shall be paid by client to consultant at any circumstances.
 - ❖ All matters related to this project shall be governed by the laws of Nepal.



D. Evaluation of Consultant's EOI Application

Evaluation of Consultant's EOI Application

Consultant's EOI application which meets the eligibility criteria will be ranked on the basis of the Ranking Criteria.

i) Eligibility & Completeness Test

Sl. No.	Criteria Title	Compliance
1	Corporate Registration	
2	Tax Clearance/Tax Return Submission	
3	VAT/PAN Registration	
4	EOI Form 1: Letter of Application	
5	EOI Form 2: Applicant's Information Form	
6	EOI Form 3: Experience (3(A) and 3(B))	
7	EOI Form 4: Capacity	
8	EOI Form 5: Qualification of Key Experts	
9	In case of a natural person or firm/institution/company which is already declared blacklisted and ineligible by the GoN, any other new or existing firm/institution/company owned partially or fully by such Natural person or Owner or Board of director of blacklisted firm/institution/company; shall not be eligible consultant.	
10	If the corruption case is being filed to Court against the Natural Person or Board of Director of the firm/institution /company or any partner of JV, such Natural Person or Board of Director of the firm/institution /company or any partner of JV shall not be eligible to participate in procurement process till the concerned Court has not issued the decision of clearance against the Corruption Charges.	
11	Each Participating JVs should have a minimum annual turnover (Best of 3 Fiscal Year of Last 7 Fiscal Years) of 0.07 billion.	
12	At least 7 years of standing for all JV partners.	
13	Formal Joint Venture Agreement Letter (in case of JV).	
14	In case of International Consulting firm, the firm should form joint venture with at least one National consulting firm.	

ii) EOI Evaluation Criteria

A. Qualification

Sl. No.	Criteria	Minimum Requirement
1	Qualification of Key Experts: Team Leader	At Least, Master's Degree in Geomatics engineering/Geoinformatics or GIS and Remote sensing or Environmental Science/ Environmental Engineering or Forestry or Natural Resource Management or Applied Earth Science urban/rural/regional Planning or Geography with GIS and Remote Sensing, Photogrammetric. Preferable Ph.D.in one of the above mentioned subjects. With GIS and Remote Sensing included in course or at least six months mandatory training in GIS and Remote sensing or having post-graduate diploma in GIS or Remote sensing as an extra academic achievement in addition to the specified master degree if GIS and RS are not included in course.
2	Experience of Key Experts: Team Leader	At least 5 projects of relevant work experience in LiDAR Mapping and Surveying field. After completion of Basic Degree .
3	Qualification of Key Experts: Deputy Team Leader	At Least, Master's Degree in Geomatics engineering/Geoinformatics or GIS and Remote sensing or

Sl. No.	Criteria	Minimum Requirement
		Environmental Science/ Environmental Engineering or Forestry or Natural Resource Management or Applied Earth Science urban/rural/regional Planning or Geography with GIS and Remote Sensing, Photogrammetric. Preferable Ph.D.in one of the above mentioned subjects. With GIS and Remote Sensing included in course or at least six months mandatory training in GIS and Remote sensing or having post-graduate diploma in GIS or Remote sensing as an extra academic achievement in addition to the specified master degree if GIS and RS are not included in course.
4	Experience of Key Experts: Deputy Team Leader	At least 4 projects of relevant work experience in related field. After completion of Basic Degree
5	Qualification of Key Experts: Senior Surveyor/ Geomatic Engineer 1	Having Completed at least Senior Surveying Course (or equivalent to senior surveying Course)/or B.E. in Geomatics Engineering/Survey engineering.
6	Experience of Key Experts: Senior Surveyor/ Geomatic Engineer 1	At least 2 projects of relevant work experience in related field after Completion of the relevant course After completion of Basic Degree
7	Qualification of Key Experts: Senior Surveyor/ Geomatic Engineer 2	Having Completed at least Senior Surveying Course (or equivalent to senior surveying Course)/or B.E. in Geomatics Engineering/Survey engineering.
8	Experience of Key Experts: Senior Surveyor/ Geomatic Engineer 2	At least 2 projects of relevant work experience in related field after Completion of the relevant course After completion of Basic Degree
9	Qualification of Key Experts: GIS /RS Expert 1	At least having BE in geomatic engineering preferable master's Degree with GIS/RS included and studied in course or having PG diploma in GIS/RS after the completion of Bachelor's Degree.
10	Experience of Key Experts: GIS /RS Expert 1	At least 2 projects of relevant work experience in related field after completion of relevant course. After completion of Basic Degree
11	Qualification of Key Experts: GIS /RS Expert 2	At least having BE in geomatic engineering preferable master's Degree with GIS/RS included and studied in course or having PG diploma in GIS/RS after the completion of Bachelor's Degree.
12	Experience of Key Experts: GIS /RS Expert 2	At least 2 projects of relevant work experience in related field after completion of relevant course. After completion of Basic Degree
13	Qualification of Key Experts: GPS Operator 1	One year Junior Surveyor Course or Diploma in Geomatics.
14	Experience of Key Experts: GPS Operator 1	At least 1 projects of relevant work experience in related field. After completion of Basic Degree
15	Qualification of Key Experts: GPS Operator 2	One year Junior Surveyor Course or Diploma in Geomatics.
16	Experience of Key Experts: GPS Operator 2	At least 1 projects of relevant work experience in related field. After completion of Basic Degree
17	Qualification of Key Experts: GPS Operator 3	One year Junior Surveyor Course or Diploma in Geomatics.
18	Experience of Key Experts: GPS Operator 3	At least 1 projects of relevant work experience in related field. After completion of Basic Degree
19	Qualification of Key Experts: GPS Operator 4	One year Junior Surveyor Course or Diploma in Geomatics.
20	Experience of Key Experts: GPS Operator 4	At least 1 projects of relevant work experience in related field. After completion of Basic Degree

Score: 40.0

B. Experience

Sl. No.	Criteria	Minimum Requirement
---------	----------	---------------------

Sl. No.	Criteria	Minimum Requirement
1	General Experience of consulting firm	5 Projects
2	Specific experience of consulting firm within last 7 years. In case of person, specific experience of the person within last 4 years.	- Airborne LiDAR Surveying/LiDAR Mapping experience of less than 2,300 sq. km. in total will not be evaluated. - Airborne LiDAR Surveying/LiDAR Mapping experience for individual projects less than 500 sq. km. will not be evaluated.
3	Similar Geographical experience of consulting firm	- Airborne LiDAR Surveying/LiDAR Mapping experience of less than 2,300 sq. km. in total will not be evaluated. - Airborne LiDAR Surveying/LiDAR Mapping experience for individual projects less than 500 sq. km. will not be evaluated.

Score: 45.0

C. Capacity

Sl. No.	Criteria	Minimum Requirement
1	Financial Capacity.[Average turnover required shall not exceed 150% of cost estimate]	0.21 billion
2	Equipment 1: Aircraft	1
3	Equipment 2: DGPS Equipment	at least 8 equipment
4	Infrastructure	LiDAR Post Processing Facility

Score: 15.0

Minimum score to pass the EOI is: 70

Note : If the corruption case is being filed to Court against the Natural Person or Board of Director of the firm/institution /company or any partner of JV, such Natural Person or Board of Director of the firm/institution /company or any partner of JV such consultant's proposal shall be excluded during the evaluation.

E. EOI Forms & Formats

E. EOI Forms & Formats

Form 1. Letter of Application

Form 2. Applicant's information

Form 3. Experience (*General, Specific and Geographical*)

Form 4. Capacity

Form 5. Qualification of Key Experts

Standard EOI Document

1. Letter of Application

(Letterhead paper of the Applicant or partner responsible for a joint venture, including full postal address, telephone no., fax and email address)

Date:

To,

Full Name of Client: _____

Full Address of Client: _____

Telephone No.: _____

Fax No.: _____

Email Address: _____

Sir/Madam,

1. Being duly authorized to represent and act on behalf of (hereinafter "the Applicant"), and having reviewed and fully understood all the short-listing information provided, the undersigned hereby apply to be short-listed by **[Insert name of Client]** as Consultant for **[Insert brief description of Work/Services]**.
2. Attached to this letter are photocopies of original documents defining:
 - a) the Applicant's legal status;
 - b) the principal place of business;
3. **[Insert name of Client]** and its authorized representatives are hereby authorized to verify the statements, documents, and information submitted in connection with this application. This Letter of Application will also serve as authorization to any individual or authorized representative of any institution referred to in the supporting information, to provide such information deemed necessary and requested by yourselves to verify statements and information provided in this application, or with regard to the resources, experience, and competence of the Applicant.
4. **[Insert name of Client]** and its authorized representatives are authorized to contact any of the signatories to this letter for any further information.¹
5. All further communication concerning this Application should be addressed to the following person,

[Person]

[Company]

[Address]

[Phone, Fax, Email]
6. We declare that, we have no conflict of interest in the proposed procurement proceedings and we have not been punished for an offense relating to the concerned profession or

¹ Applications by joint ventures should provide on a separate sheet, relevant information for each party to the Application.

Standard EOI Document

business and our Company/firm has not been declared ineligible.

7. We further confirm that, if any of our experts is engaged to prepare the TOR for any ensuing assignment resulting from our work product under this assignment, our firm, JV member or sub-consultant, and the expert(s) will be disqualified from short-listing and participation in the assignment.
8. The undersigned declares that the statements made and the information provided in the duly completed application are complete, true and correct in every detail.

Signed :

Name :

For and on behalf of (name of Applicant or partner of a joint venture):

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2. Applicant's Information Form

(In case of joint venture of two or more firms to be filled separately for each constituent member)

1. Name of Firm/Company:
2. Type of Constitution (*Partnership/ Pvt. Ltd/Public Ltd/ Public Sector/ NGO*)
3. Date of Registration / Commencement of Business (*Please specify*):
4. Country of Registration:
5. Registered Office/Place of Business:
6. Telephone No; Fax No; E-Mail Address
7. Name of Authorized Contact Person / Designation/ Address/Telephone:
8. Name of Authorized Local Agent /Address/Telephone:
9. Consultant's Organization:
10. Total number of staff:
11. Number of regular professional staff:

(Provide Company Profile with description of the background and organization of the Consultant and, if applicable, for each joint venture partner for this assignment.)

3. Experience

3(A). General Work Experience

(Details of assignments undertaken. Each consultant or member of a JV must fill in this form.)

S. N.	Name of assignment	Location	Value of Contract	Year Completed	Client	Description of work carried out
1.						
2.						
3.						
4.						
5.						
6.						
7.						

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3(B). Specific Experience

Details of similar assignments undertaken in the previous seven years

(In case of joint venture of two or more firms to be filled separately for each constituent member)

Assignment name:	Approx. value of the contract (in current NRs; US\$ or Euro) ² :
Country: Location within country:	Duration of assignment (months):
Name of Client:	Total No. of person-months of the assignment:
Address:	Approx. value of the services provided by your firm under the contract (in current NRs; US\$ or Euro):
Start date (month/year): Completion date (month/year):	No. of professional person-months provided by the joint venture partners or the Sub-Consultants:
Name of joint venture partner or sub-Consultants, if any:	Narrative description of Project:
Description of actual services provided in the assignment: Note: Provide highlight on similar services provided by the consultant as required by the EOI assignment.	

Firm's Name: _____

² Consultant should state value in the currency as mentioned in the contract

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3(C). Geographic Experience

Experience of working in similar geographic region or country

(In case of joint venture of two or more firms to be filled separately for each constituent member)

No	<i>Name of the Project</i>	<i>Location (Country/ Region)</i>	<i>Execution Year and Duration</i>
1.			
2.			
3.			
4.			
5.			
6.			
7.			

4. Capacity

4(A). Financial Capacity

(In case of joint venture of two or more firms to be filled separately for each constituent member)

Annual Turnover	
Year	Amount Currency

- Average Annual Turnover

--

(Note: Supporting documents for Average Turnover should be submitted for the above.)

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4(B). Infrastructure/equipment related to the proposed assignment³

No	Infrastructure/equipment Required	Requirements Description
1.		
2.		
3.		
4.		
5.		

³ Delete this table if infrastructure/equipment for the proposed assignment is not required.

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5. Key Experts *(Include details of Key Experts only)*

(In case of joint venture of two or more firms to be filled separately for each constituent member)

SN	Name	Position	Highest Qualification	Work Experience (in year)	Specific Work Experience (in year)	Nationality
1						
2						
3						
4						
5						

(Please insert more rows as necessary)



Expression of Interest (EOI)

Title of Consulting Service:
Procurement of International Consulting
Services -Generation of High-Resolution DEM
and Ortho photo Map with application of
Airborne LiDAR Surveying and Mapping
SD/ICS/QCBS/01/2081/082

Method of Consulting Service: *International*

Issued By:
Survey Department
Minbhwan, Kathmandu

April 23, 2025

Expression of Interest (EOI)

Title of Consulting Services
Procurement of International Consulting
Services for Generation of High Resolution
DEM and Ortho photo Map with application of
Airborne LiDAR Surveying and Mapping
SD/ICS/QCBS/01/2081/082

Method of Consulting Service
International

Project Name: *Procurement of International Consulting Services For
Generation of High Resolution DEM and Ortho photo Map with
application of Airborne LiDAR Surveying and Mapping*

EOI: *SD/ICS/QCBS/01/2081/082*

Office Name: *Survey Department*

Office Address: *Minbhawan, Kathmandu*

Issued on: *April 23, 2025*

Financing Agency: Government Budget



Abbreviations

CV	-	Curriculum Vitae
DO	-	Development Partner
EA	-	Executive Agency
EOI	-	Expression of Interest
GIID	-	Geographic Information Infrastructure Division
GON	-	Government of Nepal
PAN	-	Permanent Account Number
PPA	-	Public Procurement Act
PPR	-	Public Procurement Regulation
TOR	-	Terms of Reference
VAT	-	Value Added Tax



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A. Request for Expression of Interest

Government of Nepal
Ministry of Land Management, Cooperatives and Poverty Alleviation
Survey Department

Date: April 23, 2025

Name of Project: Procurement of International Consulting Services for Generation of High Resolution DEM and Ortho photo Map with application of Airborne LiDAR Surveying and Mapping

1. Government of Nepal (GoN) has allocated fund **toward the cost of Procurement of International Consulting Services for Generation of High-Resolution DEM and Ortho photo Map with application of Airborne LiDAR Surveying and Mapping** and intends to apply a portion of this **fund** to eligible payments under the Contract for which this Expression of Interest is invited for **International consulting service**.
2. The *Survey Department* now invites Expression of Interest (EOI) from eligible consulting firms ("consultant") to provide the following consulting services: **Procurement of International Consulting Services for Generation of High-Resolution DEM and Ortho photo Map with application of Airborne LiDAR Surveying and Mapping**.
3. Interested eligible consultants may obtain further information at the address **Survey Department, Minbhawan, Kathmandu** during office hours before **submission deadline** or visit e-GP system www.bolpatra.gov.np/egp.
4. Expression of Interest may be submitted by a sole firm or a joint venture of consulting firms and the maximum number of partners in JV shall be limited to three. The consulting firm with one JV shall not form JV with other consulting firms. In case of International Consulting firm, the firm should form joint venture with at least one National consulting firm. In case of JV, name, address and other detail of all JV partners and name of leading firm should be clearly mentioned.
5. Expressions of interest shall be delivered online through e-GP system www.bolpatra.gov.np/egp on or before 12:00 hours on **May 08, 2025** EOI shall be opened in 13:00 hours on **May 08, 2025** at the office of Survey Department, Minbhawan Kathmandu.
6. In case the last date of obtaining and submission of the EOI documents happens to be a holiday, the next working day will be deemed as the due date but the time will be the same as stipulated.
7. EOI will be assessed based on **Qualification 40 %, Experience 45 %, and Capacity 15%** of consulting firm and key personnel on the basis of the submitted documents and proof. Based on evaluation of EOI, only shortlisted firms will be invited to submit technical and financial proposal through a request for proposal.
8. Minimum score to pass the EOI is **70%**.
9. A Consultant will be selected in accordance with the **Quality Cost Based Selection (QCBS)** method and the minimum score to pass the EOI is **70%**.
10. Documents other than Nepali and English shall be translated to English and attested from notary.
11. Eligible consultants will be selected for RFP according to the provision of Public Procurement Act, 2063 and Public Procurement regulation, 2064.



B. Instructions for submission of Expression of Interest

1. Expression of Interest may be submitted by a sole firm or a joint venture of consulting firms and the maximum number of partners in JV shall be limited to three. The consulting firm with one JV shall not form JV with other consulting firms. In case of International Consulting firm, the firm should form joint venture with at least one National consulting firm.
2. A consulting firm shall not submit EOI more than once, either individually or with JV.
3. Interested consultants must provide information indicating that they are qualified to perform the services (*descriptions, organization and employee and of the firm or company, description of assignments of similar nature completed in the last 7 years and their location, experience in similar conditions, general qualifications and the key personnel to be involved in the proposed assignment*).
4. This expression of interest is open to all eligible.
5. In case, the applicant is individual consultant, details of similar assignment experience, their location in the previous 4 years and audited balance sheet and bio data shall be considered for evaluation.
6. The assignment has been scheduled for a period of **21 months**. Expected date of commencement of the assignment is **[insert date]**.
7. A Consultant will be selected in accordance with the QCBS method.
8. Expression of Interest should contain following information:
 - (i) A covering letter addressed to the representative of the client on the official letter head of company duly signed by authorized signatory.
 - (ii) Applicants shall provide the following information in the respective formats given in the EOI document:
 - *EOI Form: Letter of Application (Form 1)*
 - *EOI Form: Applicant's Information (Form 2)*
 - *EOI Form: Work Experience Details (Form 3(A), 3(B) & 3(C))*
 - *EOI Form: Capacity Details (Form 4)*
 - *EOI Form: Key Experts List (form 5).*
9. Applicants may submit additional information with their application but shortlisting will be based on the evaluation of information requested and included in the formats provided in the EOI document.
10. The Expression of Interest (EOI) document must be duly completed and submitted **by electronically only using the forms and instructions provided by the system**.
11. The completed EOI document must be submitted on or before the date and address mentioned in the **"Request for Expression of Interest."** In case the submission falls on public holiday the submission can be made on the next working day. Any EOI Document received after the closing time for submission of proposals shall not be considered for evaluation.



C. Objective of Consultancy Services or Brief TOR
Terms of Reference (ToR)
For

Procurement of International Consulting Services - Generation of High Resolution DEM and Ortho photo Map with application of Airborne LiDAR

A. Background

Survey Department, Ministry of Land Management, Cooperatives and Poverty Alleviation, Government of Nepal is the national mapping organization, is mandated to provide digital topographical mapping and other related base map services as well as information. Geographic Information Infrastructure Division (GIID), under Survey Department, was initially set up as Census Mapping Project in 1998 AD, which was later upgraded to National Geographic Information Infrastructure Project (NGIIP) later in 2002 AD. The topographic base maps prepared from the FINIDA and JICA funded topographic mapping projects were also updated and integrated to National Topographic Database (NTDB) under the project. Later upgraded to Geographic Information Infrastructure Division (GIID) in 2015 AD, the division is now one of four divisions under Survey Department. It mainly acts as the National Spatial Information Infrastructure. Since then, it has been contributing as the national data clearinghouse, that offers spatial data and services through National Geoportal. Since its upgrade, one of the major achievement has been successful commencement of Airborne LiDAR survey of 20,000 square kilometers area of Western Terai of Nepal.

LiDAR, abbreviation for Light Detection and Ranging, is a remote sensing method that uses light in the form of a pulsed laser to measure ranges (variable distances) to the Earth. Airborne LiDAR consists of high frequency laser scanner that records the time difference between the emission of the laser pulses and the reception of the reflected signal from the environment. These light pulses combined with other data recorded by the airborne system generate precise, three-dimensional information about the shape of the Earth and its surface characteristics. LiDAR creates highly accurate and detailed models of the earth's surface with horizontal and vertical accuracy down to centimeters level. Another major component of the mission is to acquire aerial photographic images of the region to produce high resolution orthophotograph. These products are the backbone to extraction of details at higher accuracy level for large scale mapping.

LiDAR systems allow mapping professionals to examine both natural and manmade environments with accuracy, precision, and flexibility. LiDAR has been used in variety of platforms and implication including digital elevation models for use in geographic information systems, to assist in emergency response operations, and in many other applications.

B. Objective of the Service

The major objective of the assignment is to prepare

- i. Contour at 0.5m interval,
- ii. Digital Elevation Model (DEM) at 1m resolution,
- iii. Digital Surface Model (DSM) at 1m resolution,
- iv. Orthophoto at resolution (GSD) of 0.10 m, of project area,
As prime outputs and
- v. Control Points Establishment by Differential GNSS
As supplementary output.



C. Specification of LiDAR Surveying and Mapping

i. **Coverage**

Area of interest (AOI) for the project will be marked in the map (Appendix A) and in digital shape file format will be provided later. Except for Nepalese International Border, area of interest (AOI) should be buffered by 150 meters.

ii. **LiDAR Point Cloud Density**

LiDAR Point Cloud Density must be at least **10 points/m²** (*except in wáter bodies*), however considering terrain and other conditions, point density may be increased to maintain desired accuracy for acieving the quality end products whenever required.

iii. **Ground sample distance (GSD) of the aerial photographs**

The digital aerial photographic images should be captured with a GSD of 0.08 m. or better at nadir.

iv. **Photographic Coverage**

The forward overlap (forelap) between successive exposures in each run should be minimum 70 percent and the lateral overlap (sidelap) between adjacent strips should be minimum 30 percent. However, these values may be increased to maintain desired accuracy.

v. **Fundamental Spatial Accuracy**

The vertical accuracy or the **root mean square error (RMSE)** of the end products should be within a range of ± 0.25 m or better and the horizontal accuracy or the **root mean square error (RMSE)** of orthophoto should not exceed three times the pixel size.

vi. **Coordinate Reference System (CRS):**

- **Vertical Coordinate System**

Primary vertical coordinate system will be elipsoidal height base don WGS 84. Data that are related to elevation should be adjusted to local height datum using geoid model.

- **Horizontal Coordinate System**

The primary horizontal coordinate system will be UTM projection based on WGS84. The deliverables should also be transformed to National Coordinate System.

vii. **Ground Control System**

- Maximum distance between the Reference GNSS station on the ground and airborne GNSS units must not exceed 25 kilometers.
- All control survey data used or derived from this assignment must be supplied to ensure independent Quality Assurance (QA) of the survey operations. It is therefore essential that all primary ground stations should be visible in photographs.
- The primary ground control and check point surveys must be referenced to survey control marks with geodetic control points (in terms of coordinates and height) demarcated by survey department.



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- Elevation data must be validated and corrected for systematic errors to ensure accuracy specifications are met. Documentation must describe how this has been achieved.

viii. LiDAR Data acquisition details

- A Draft Pre-flight Agreement with provision whereby LiDAR data over the Study Area must not be collected during any period, e. g. flood, adverse weather etc., where extent of LiDAR ground returns in any part of the Study Area is likely to be significantly compromised.
- A Draft Pre-Flight Agreement will include provision whereby the Principal is notified of each proposed LIDAR collection flight with sufficient notice to enable consultation between Principal and the contractor to determine if data capture by the Contractor should proceed.
- Collections in high relief terrain are expected to require greater overlap. Any data with gaps between the geometrically usable portions of the swaths will be rejected.
- The spatial distribution of geometrically usable points is expected to be uniform and free from clustering. In order to ensure consistent data densities throughout the project area:

Environmental conditions for data capture:

- Cloud and fog free between the vehicle and the ground.
- Floodplain/wetland data must be captured during times of base flow and outside of significant surface inundation due to natural events and/or regulated environmental flows.
- Details of the vehicle (- helicopter or aeroplane), navigation and mission planning activities for LIDAR and digital photography acquisition. Include details of whether the photography and LIDAR will be acquired during the same mission (i.e. from the same vehicle - helicopter or aeroplane) or from separate missions. If separate missions are required to satisfy the respective resolution requirements for LIDAR and photography (eg. Due to flying height constraints) then specify alternative mission scenarios based on a sensor type and platform.

ix. Orthophoto

- RGB and single band NIR orthophoto at resolution of 0.1 m, should be generated by stereoplotting.
- Tiled delivery, as per Data Supply Specifications.

x. Digital Surface Model (DSM) (orthometric)

- Digital Surface Model (DSM) at a resolution of 1m grid.
- The DSM should be generated from the “first return” LiDAR mass point data. This will include ground and non-ground points such as vegetation and buildings.
- The DSM generation should employ a Point to TIN and TIN to Raster process with Natural Nearest Neighbour interpolation.



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- Void areas (i.e., areas outside the project boundary but within any tiling scheme) shall be coded using a unique “NODATA” value.
- Tiled delivery, as per Data Supply Specifications.

xi. Digital Terrain Model (DTM)(orthometric)

- 1m grid bare earth Digital Elevation Model (DEM).
- The DEM should be generated from the LiDAR mass point data classified as “Ground” only, so that it defines the “bare earth” ground surface.
- The DEM generation should employ a Point to TIN and TIN to Raster process with Natural Nearest Neighbour interpolation.
- Void areas (i.e., areas outside the project boundary but within any tiling scheme) shall be coded using a unique “NODATA” value
- Tiled delivery, as per Data Supply Specifications.

xii. Differential GNSS Data collection

- GNSS data for all base station occupations in excess of 4 hours is to be provided in RINEX version (Receiver Independent Exchange Format) 2.10 or recent format.
- GNSS observation log sheets which include the following details:
 - a. Survey mark id
 - b. Occupation time & date
 - c. Antenna height measurements
 - d. Instrument /antenna types & serial numbers.
- The GNSS observation log sheets should be provided in pdf format or Excel spreadsheet if data is captured digitally.

xiii. Metadata

- For each supplied data product a complete metadata statement consistent with the ISO Standard.
- Metadata must be provided with final deliveries.
- The job will not be signed off by Contract Authority until the metadata is satisfactorily supplied.

xiv. Spatial Accuracy Validation

- The fundamental vertical accuracy of the point cloud dataset must be determined with check points located only in open, relatively flat terrain, where there is a very high probability that the sensor will have detected the ground surface.
- The vertical accuracy of the point cloud dataset is to be tested using a TIN surface constructed from bare-earth LiDAR points compared against ground survey check points.
- Check points are to be surveyed independently of any LiDAR GNSS operations.



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- The number of check points (locations) is dependent on the extent of the survey. The following strategy should be used as a guide:
 - a. Check points must be established to adequately cover the full extent of the survey area, and be representative of the project area landscape.
 - b. A minimum of 1 point per 50km² in 7 km * 7 km grid in different vegetation types and distributed evenly over the Project area.
- The proposed check point survey design must be submitted with the quotation, and approved by the Contract Authority prior to implementation. Acceptance of the post-survey spatial accuracy report discussed above is dependent on the quality, number and distribution of these check points.
- In the above circumstances a “compiled to meet” statement of horizontal accuracy at 95 percent confidence should be reported.

xv. **Intermediate Deliverables**

- Deliverable 1: mission flight plans in ESRI shape file format.
- Deliverable 2: Report on planned ground control, check points – including details of connections to national control network - in Word and ESRI Shape files.
- Deliverable 3: Report on planned check points to be collected by the Consultant. At least 1 check points must lie within 7 km X 7 km - Word and ESRI Shape files.

xvi. **Final Deliverables**

LiDAR Deliverables

- Deliverable 1: Raw LiDAR (point cloud) survey data comprising all returns (1st, 2nd, etc., and last return), representing all ground and non-ground returns and including overlapping swathes - in ASPRS LAS (LAS v 1.2 or stable), file format X, Y, Z coordinates, return intensity value, time and date for each point. All points retained, and classified as per LAS v 1.2 or stable (See: <http://www.lasformat.org/>)
- Trajectory files providing information about the scanner's position while capturing the point cloud.
- Deliverable 2: Digital Surface Model (DSM) and Digital Terrain Model (DTM) of 1m grid resolution - DEM should be supplied in geoTiff format in original WGS 84 and National Coordinate System.
- Deliverable 3: Contour of 0.5 m interval - Contour should be supplied in ESRI shape file original WGS 84 and National Coordinate System..

Orthophoto

- Deliverable 4: Near-natural colour-balanced digital photography for photogrammetry application. Georeferenced, but not ortho-rectified. 0.10 m. ground sample distance (GSD) - Photographs with GPS INV data, Campfile and IMG/TIFF/GEOTIFF format - Red, Green & Blue bands.
- Deliverable 5: Ortho rectified multispectral digital photograph with 0.10 m. GSD. – GEOTIFF Red, Green, Blue.



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- Deliverable 6: Colour balanced mosaic with 0.10 m. GSD - Red, Green and Blue bands.

Documents

- Flight Data Storage (FDS) report: A report shall be delivered with the raw data (on FDS) giving the following information:
 1. Project name.
 2. Number of images.
 3. FDS metadata report
 4. Year(s), month(s) and day(s) of raw data.
 5. Camera type and Camera Calibration Information.
 6. Run number and flight direction.
 7. Aircraft type and identification.
 8. Names of pilot(s), navigator and photographer.
 9. Start and end time for each run in local time.
 10. Computed altitude above mean sea level (true altitude).
 11. GSD of output images.
 12. Weather conditions – cloud type, degree of haze, turbulence etc.
 13. General comment on quality.
- Output Images Report: A report shall be delivered with the output images giving the following information:
 1. Project name
 2. Number of images
 3. Year(s), month(s) and day(s) of raw data.
 4. Camera type and Camera Calibration Information.
 5. Strip number and flight direction.
 6. Aircraft type and identification.
 7. Names of pilot(s), navigator and photographer.
 8. Start and end time for each run in local time.
 9. Computed altitude above mean sea level (true altitude).
 10. Type and GSD of output images.
 11. Date of processing
 12. Operator of Post Processing
 13. Weather conditions - cloud type, degree of haze, turbulence etc.
 14. General comment on quality
- Other material to be delivered
 1. Plot of planned flight missions
 2. Plot of finished flight missions
 3. Aerial survey report
- Point cloud classification report
 1. Software used
 2. Point cloud classification rules
 3. Detail on manual classification
 4. Point cloud classification accuracy assessment data
- Detail Final report with all necessary details.

xvii. **Project Period:**
Total time duration of project is 21 months.

xviii. **Team Composition of Key Experts**

S.N.	Position	Qualification	Experience	Additional Qualification	Number of Person
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S.N.	Position	Qualification	Experience	Additional Qualification	Number of Person
1.	Team Leader	At Least, Master's Degree in Geomatics engineering/Geoinformatics or GIS and Remote sensing or Environmental Science/ Environmental Engineering or Forestry or Natural Resource Management or Applied Earth Science urban/rural/regional Planning or Geography with GIS and Remote Sensing, Photogrammetric. Preferable Ph.D.in one of the above mentioned subjects.	At least 5 projects of relevant work experience in LiDAR Mapping and Surveying field. After completion of Basic Degree .	With GIS and Remote Sensing included in course or at least six months mandatory training in GIS and Remote sensing or having post-graduate diploma in GIS or Remote sensing as an extra academic achievement in addition to the specified master degree if GIS and RS are not included in course.	1
2.	Deputy Team Leader	At Least, Master's Degree in Geomatics engineering / Geo-informatics or GIS and Remote sensing or Environmental Science/ Environmental Engineering or forestry or Natural Resource Management or Applied Earth Science urban/rural/regional Planning or Geography with GIS and Remote Sensing, Photogrammetric. Preferable Ph.D.in one of the above mentioned subjects.	At least 4 projects of relevant work experience in related field. After completion of Basic Degree	With GIS and Remote Sensing included in course or at least six months mandatory training in GIS and Remote sensing or having post-graduate diploma in GIS or Remote sensing as an extra academic achievement in addition to the specified master degree if GIS and RS are not included in course.	1
3.	Senior Surveyor/ Geomatic Engineer	Having Completed at least Senior Surveying Course (or equivalent to senior surveying Course)/or B.E. in Geomatics Engineering/Survey engineering.	At least 2 projects of relevant work experience in related field after Completion of the relevant course After completion of Basic Degree	-	2
4.	GIS /RS Expert	At least having BE in geomatic engineering preferable master's Degree with GIS/RS included and studied in course or having PG diploma in GIS/RS after the completion of Bachelor's Degree.	At least 2 projects of relevant work experience in related field after completion of relevant course. After completion of Basic Degree	-	2
5.	GPS Operator	One year Junior Surveyor Course or Diploma in Geomatics.	At least 1 projects of relevant work experience in related field. After completion of Basic Degree	-	4

xix. **Non Key Expert**

6.	Field Enumerator	At least intermediate level.		-	4
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D. LAB Setup:

To establish a fully equipped lab for LiDAR data processing and analysis, the bidder is required to provide the necessary hardware and software as outlined. The lab setup will include **four high-performance workstations** to handle complex geospatial data efficiently. Additionally, the lab will feature a comprehensive suite of ArcGIS software licenses to support a wide range of geospatial applications, including 3D analysis, data validation, geostatistical modeling, and advanced mapping capabilities. The bidder must ensure that all software licenses are genuine, correctly installed, and fully compatible with



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the workstations provided. The table below outlines the specific components required for the lab setup:

Category	Item	Quantity
Workstations	High-Performance Workstation	4
Processing and Mapping Software	ArcGIS Pro Professional Plus (64-bit) – Perpetual License (or equivalent). The software version released should be of year 2024 or latest. The software must include the following extensions:	1
	3D Analyst	1
	Data Reviewer	1
	Geostatistical Analyst	1
	Image Analyst	1
	Locate XT	1
	Network Analyst	1
	Publisher	1
	Spatial Analyst	1
	Workflow Manager	1

Note: The lab is designed to meet the demands of advanced geospatial analysis and mapping projects. The bidder must ensure timely delivery, installation, and proper configuration of all components for the lab to be operational.

Specification of Workstation:

SN	Components	Requirement
1	CPU	AMD Ryzen Threadripper 7960X 24-Core, 48-Thread 5.3GHz Max
2	RAM	128GB ECC RDIMM - Lifetime Warranty
3	Video Card	Nvidia RTX 4060 Ti 16GB GDDR6 4352 CUDA Cores
4	CPU Cooler	Threadripper Water Cooler - 3x120mm Fans - Sealed & Maintenance Free - 1 Yr Warranty - Highest Performance Option
5	Storage	PCIe NVMe SSD(s) High Performance SSD – 4,000GB - Up to 7,250 MBps - 1 Yr Warranty - NVMe PCIe 4.0
6	OS	Factory Enabled Genuine Windows 11 professional 64 bit
7	Warranty	1-Years Parts, 1-Year Advanced Parts Replacement
8	Accessories	Mouse and Keyboard
9	Monitor	27” UHD 4K



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E. Training

Training Should be provided as proposed by the survey department. The table provides a clear summary of the objectives, number of participants, and the costs to be covered for both the Course on Lidar Scanning (1 month) and the Exposure Visit (1 week), ensuring bidders have a detailed understanding of the requirements for their proposals.

Training Component	Objective	Number of Participants	Costs to be Covered
Course on Lidar Scanning and Data Processing (1 month)	To equip participants with technical knowledge and practical skills in Lidar technology as specified in the course below.	5	- Course Fee - Round-Trip Travel Costs - Accommodation Rent - Daily Allowance (meals and local transport) - Travel Insurance
Exposure Visit (1 week)	To provide practical insights and exposure to best practices in Lidar application for urban settings.	4	- Round-Trip Travel Costs - Hotel Accommodation - Daily Allowance (meals and local transport) - Travel Insurance

Course Overview:

This comprehensive course, offered by an International Institute or Academia certified in LiDAR data acquisition and processing, must cover the principles of LiDAR technology with a strong emphasis on data acquisition and processing. The course should provide a thorough understanding of LiDAR technology, including its principles, data acquisition techniques, and processing methods. Participants must gain hands-on experience with LiDAR datasets, using Terrasolid software and other industry-standard tools. They should learn how to apply various processing algorithms for generating and analyzing orthophotos and Digital Elevation Models (DEMs), and explore real-world applications in fields such as geography, environmental science, urban planning, and forestry.

Course Objectives:

- Understand the fundamental principles of LiDAR technology.
- Acquire skills in LiDAR data collection and processing.
- Generate and analyze orthophotos and Digital Elevation Models (DEMs).
- Utilize software tools for LiDAR data visualization and analysis.
- Interpret LiDAR data for various applications.

Modules:

Module 1: Introduction to LiDAR

- **Lecture 1: Fundamentals of LiDAR Technology**
 - Cover principles of LiDAR.
 - Explain types of LiDAR systems (Terrestrial, Airborne, Mobile).
- **Lecture 2: History and Development of LiDAR**
 - Discuss the evolution of LiDAR technology.
 - Highlight key milestones and advancements.



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- **Practical Session:** Provide an overview of LiDAR equipment and software.

Module 2: LiDAR Data Acquisition

- **Lecture 1: Planning LiDAR Surveys**
 - Discuss survey design considerations.
 - Explain flight planning for airborne LiDAR.
- **Lecture 2: Data Collection Methods**
 - Teach terrestrial and mobile LiDAR scanning techniques.
 - Cover airborne LiDAR data collection.
- **Lecture 3: Best Practices in Data Acquisition**
 - Ensure quality assurance and control.
 - Discuss safety and regulatory considerations.
- **Practical Session:** Conduct a LiDAR survey (fieldwork).

Module 3: LiDAR Data Processing

- **Lecture 1: Introduction to LiDAR Data Processing**
 - Provide an overview of processing workflows.
 - Introduce software tools for processing.
- **Lecture 2: Pre-processing of LiDAR Data**
 - Cover data import and format conversion.
 - Teach noise reduction and filtering techniques.
- **Lecture 3: Point Cloud Processing with Terrasolid**
 - Teach classification of LiDAR points.
 - Demonstrate generating Digital Elevation Models (DEMs).
- **Lecture 4: Orthophoto Generation**
 - Explain integration of LiDAR and imagery.
 - Demonstrate creating orthophotos from aerial imagery.
- **Practical Session:** Conduct hands-on LiDAR data processing using Terrasolid software.

Module 4: Applications of LiDAR

- **Lecture 1: Environmental Applications**
 - Teach forest mapping and biomass estimation.
 - Cover flood modeling and watershed analysis.
- **Lecture 2: Urban and Infrastructure Applications**
 - Teach urban planning and 3D city modeling.
 - Cover transportation and infrastructure management.
- **Practical Session:** Provide case studies and project work.

Module 5: Advanced LiDAR Processing Techniques

- **Lecture 1: Advanced Features of Terrasolid**
 - Teach feature extraction and 3D modeling.
 - Cover automation and scripting in Terrasolid.
- **Lecture 2: Machine Learning and LiDAR**
 - Integrate machine learning for LiDAR data analysis.



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- Demonstrate automated feature extraction.
- **Lecture 3: Multi-sensor Data Fusion**
 - Teach combining LiDAR with other geospatial data (e.g., imagery, GPS).
 - Discuss benefits and challenges of data fusion.
- **Practical Session:** Conduct advanced processing exercises.

Note: All expenses for travel, insurance, accommodation, and the course fee for Five participants must be covered by the consultant and ensures participants can fully engage in the course focusing on acquiring valuable skills and knowledge in LiDAR technology.

F. Reporting Stages

Reports should be provided at various stages for regular monitoring of the project and engagement of stakeholders. The Consultants shall submit the following reports in the specified timeframe as mentioned below:

Report Type	Content	Submission Timeline	Copies Required
Inception Report	- Feasibility analysis and survey plan. - Identification of priority areas and logistical considerations.	Within 15 days from the date of contract.	3 hard copies with softcopy.
Interim Progress Reports	- Updates on data collection milestones and preliminary results. - Identification of challenges and corrective measures.	Every quarter.	3 hard copies with softcopy.
Final Report	- Comprehensive dataset delivery (including DEMs, DSMs, and classified point clouds). - Summary of findings, applications, and recommendations for utilization. - Draft report submission required before the final report.	Draft: Before Final Report Final: Upon draft approval.	6 hard copies with softcopy.
Post-Survey Review	- Evaluation of project outcomes and lessons learned. - Suggestions for future updates or maintenance of datasets.	After the completion of the survey.	2 hard copies with softcopy.

G. Payment Schedule

- **First Payment:** Ten (10) percent of the contract price shall be paid after submission of Satisfactory Inception report acceptable to the Client
- **Second Payment:** Fifteen (15) percent of the contract price shall be paid within Fifteen (15) days of submission of Satisfactory DGPS field survey report and Flight Plan Report acceptable to the client.
- **Third Payment:** Ten (10) percent of the contract price shall be paid after supply and installation of high performance workstations and Software.



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- **Fourth Payment:** Twenty (20) percent of the contract price shall be paid within Thirty (30) days of submission of Satisfactory LiDAR data capture report and raw data acceptable to the client
- **Fifth Payment:** Twenty (20) percent of the contract price shall be paid within Thirty (30) days of submission of Satisfactory Draft Deliverables on the assignments as per TOR, Technical Specification acceptable to the client.
- **Final Payment:** Twenty Five (25) percent of Contract Price shall be made within thirty (30) days following the Client's written confirmation of acceptance of the Final Deliverables, provided that such deliverables are submitted in full compliance with the Terms of Reference, technical specifications, and all comments and recommendations made during the review of the draft submissions.
- Note:
 - ❖ All Payment shall be made in Nepalese Currency.
 - ❖ No additional Charges, taxes or penalty shall be paid by client to consultant at any circumstances.
 - ❖ All matters related to this project shall be governed by the laws of Nepal.



D. Evaluation of Consultant's EOI Application

Consultant's EOI application which meets the eligibility criteria will be ranked on the basis of the Ranking Criteria.

i) Eligibility & Completeness Test	Compliance
Copy of Registration of the company/firm	
At least 7 years of standing for all JV partners.	
In case of International Consulting firm, the firm should form joint venture with at least one National consulting firm.	
Each Participating JVs should have a minimum annual turnover (<i>Best of 3 Fiscal Year of Last 7 Fiscal Years</i>) of 0.07 billion.	
VAT/PAN Registration (<i>for National consulting firm only</i>)	
Tax Clearance/Tax Return Submission/Letter of Time Extension for Tax Return Submission 2080/81 (<i>for National consulting firm only</i>)	
In case of a natural person or firm/institution/company which is already declared blacklisted and ineligible by the GoN, any other new or existing firm/institution/company owned partially or fully by such Natural person or Owner or Board of director of blacklisted firm/institution/company; shall not be eligible consultant.	
Formal Joint Venture Agreement Letter (in case of JV).	
EOI Form 1: Letter of Application	
EOI Form 2: Applicant's Information Form	
EOI Form 3: Experience (3(A) and 3(B))	
EOI Form 4: Capacity	
EOI Form 5: Qualification of Key Experts	

ii) EOI Evaluation Criteria					Insert Minimum Requirement if Applicable		Score [Out of 100%]
A. Qualification (40 %)							40
Qualification and Experience of Key Experts							[Client should insert in between: [30 % to 50 %]
SN	Team Composition	Nos	Qualification (per expert)	Experience (per expert)	Total Marks (per expert)	Total marks	
1	Team Leader	1	2.7	6.3 (<i><5 projects = 0 marks, 5 projects =70%, 6 projects = 80%, 7 projects =90%, 8 or more =100%) of total marks</i>	9	9	
2	Deputy Team Leader	1	2.1	4.9 (<i><4 projects = 0 marks, 4 projects =70%, 5 projects = 80%, 6 projects =90%, 7 or more =100%) of total marks</i>	7	7	
3	Senior Surveyor/ Geomatic Engineer	2	1.2	2.8 (<i><2 projects = 0 marks, 2 projects =70%, 3 projects = 80%, 4 projects =90%, 5 or more =100%) of total marks</i>	4	8	
4	GIS /RS Expert	2	1.2	2.8 (<i><2 projects = 0 marks, 2 projects =70%, 3 projects = 80%, 4 projects =90%, 5 or more =100%) of total marks</i>	4	8	
5	GPS Operator	4	0.6	1.4 (<i><1 projects = 0 marks, 1 project =70%, 2 projects = 80%, 3 projects =90%, 5 or more =100%) of total marks</i>	2	8	

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Total	40	
B. Experience (45 %)		45
1. General Experience of consulting firm. (<5 projects = 0 marks, 5 projects =70%, 6 projects = 80%, 7 projects =90%, 8 or more =100%) of total marks		5
2. Specific experience of consulting firm within last 7 years. - Specific work experience (Airborne LiDAR Surveying/LiDAR Mapping experience ≥3200 Sq. Km.)- 100% - Specific work experience (Airborne LiDAR Surveying/LiDAR Mapping experience ≥2900 Sq. Km.<3200 Sq. Km.)- 90% - Specific work experience (Airborne LiDAR Surveying/LiDAR Mapping experience ≥2600 Sq. Km.<2900 Sq. Km.)- 80% - Specific work experience (Airborne LiDAR Surveying/LiDAR Mapping experience ≥2300 Sq. Km.<2600 Sq. Km.)- 70% - Airborne LiDAR Surveying/LiDAR Mapping experience of less than 2,300 sq. km. in total will not be evaluated. - Airborne LiDAR Surveying/LiDAR Mapping experience for individual projects less than 500 sq. km. will not be evaluated.		30
3 Similar Geographical experience of consulting firm in Airborne LiDAR Surveying/LiDAR Mapping experience of National/International consulting firm In Nepal within last 7 years. - Specific work experience (Airborne LiDAR Surveying/LiDAR Mapping experience ≥3200 Sq. Km.)- 100% - Specific work experience (Airborne LiDAR Surveying/LiDAR Mapping experience ≥2900 Sq. Km.<3200 Sq. Km.)- 90% - Specific work experience (Airborne LiDAR Surveying/LiDAR Mapping experience ≥2600 Sq. Km.<2900 Sq. Km.)- 80% - Specific work experience (Airborne LiDAR Surveying/LiDAR Mapping experience ≥2300 Sq. Km.<2600 Sq. Km.)- 70% - Airborne LiDAR Surveying/LiDAR Mapping experience of less than 2,300 sq. km. in total will not be evaluated. - Airborne LiDAR Surveying/LiDAR Mapping experience for individual projects less than 500 sq. km. will not be evaluated.		10
C. Capacity (15%)		15
Financial Capacity (Average Annual Turnover of Best of 3 Fiscal Year of Last 7 Fiscal Years) (0.21 billion to 0.25 billion: 3 marks, >0.25 billion to 0.30 billion:4 marks, >0.30 billion:5 marks)		5
Availability of Software/equipment related to the proposed		10
Aircraft * 1		5
LiDAR Post Processing Facility		3
DGPS Equipment (at least 8 equipment) (1.5 marks for 8 equipment, 2 marks for more than 8 equipment.)		2

Note:

- The Consulting firm shall submit updated signed Curriculum vitae of each proposed experts including their commitment of availability in the proposed project. Commitment letter must contain Signature, project name, mobile number and residential number. Curriculum vitae and commitment letter should be based on standard format. CV of Government Officials will not be entertained.
- The Consulting firm should provide letters of experiences of general experiences from the client.
- EOI shall be evaluated on the basis of information and documents duly provided by the Consultant. Information must be supported by relevant evidences such as academic



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- qualification certificates, experience letters and training certificate.
4. Experts having minimum academic qualification will be allocated 70% marks for qualification and having preferred degree will be allocated 100% marks for qualification. Evaluation shall be conducted on the basis of academic documents and experience documents submitted by the consulting firm. If proposed experts failed to meet minimum academic qualification requirement for respective position, marks for qualification and experience shall be zero. And if proposed experts failed to meet minimum experience requirement then marks for experience shall be zero. Correctness and completeness of the documents is sole responsibility of the consultant/expert.
 5. The consulting firm must provide a letter of assurance that all the information on CVs submitted are true and truly signed by the respective professional himself/herself.
 6. The consulting firm must submit LiDAR Data Processing Software license confirmation certificates, provider's official letter/or authorization letter, bills/vouchers, software licenses confirmation key and/or valid hiring letter (with valid bills, vouchers, software license confirmation certificates etc.) to get allocated mark mentioned in the evaluation criteria for LiDAR Data Processing Facility. Consulting firm must submit valid certificates, bill, valid hiring letter (bills and other supporting documents) for possession of at least 1(one) LiDAR aircraft and at least 8 DGPS instruments. Without Bill/Vouchers or hiring letter (without proof of ownership of the hiring authority) will not be entertained.
 7. In case of JV all partners combined should meet the requirements mentioned in the evaluation criteria.



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Commitment letter of Key Experts/Other Experts to be involved in the proposed service (to be attached with CV):

To,

**The Director General,
Survey Department
Min Bhawan, Kathmandu, Nepal**

Subject: **Commitment Letter**

Dear Sir,

I[insert the Name of the Key Expert] would like to express my commitment, that I have provided my genuine documents(academic and experience) and duly signed my CV to be available to provide my service in the capacity of[insert position] for the Procurement of International Consulting Services For Generation of High Resolution DEM and Ortho photo Map with application of Airborne LiDAR Surveying and Mapping (SD/ICS/QCBS/01/2081/082).

I will be responsible for undertaking the above task assigned to me as a part of this project.

Your's Sincerely

Signature

Name of the Key Expert

Position:

Date:

Mobile Number:

email ID:



E. EOI Forms & Formats

Form 1. Letter of Application

Form 2. Applicant's information

Form 3. Experience (*General, Specific and Geographical*)

Form 4. Capacity

Form 5. Qualification of Key Experts

A handwritten signature in black ink, located at the bottom left of the page. The signature is stylized and appears to be a cursive script.

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1. Letter of Application

(Letterhead paper of the Applicant or partner responsible for a joint venture, including full postal address, telephone no., fax and email address)

Date:

To,
Survey Department
Minbhawan, kathmandu

Full Address of Client: _____

Telephone No.: _____

Fax No.: _____

Email Address: _____

Sir/Madam,

1. Being duly authorized to represent and act on behalf of (hereinafter "the Applicant"), and having reviewed and fully understood all the short-listing information provided, the undersigned hereby apply to be short-listed by **Survey Department** as Consultant for **Generation of High-Resolution DEM and Ortho photo Map with application of Airborne LiDAR Surveying and Mapping**.
2. Attached to this letter are photocopies of original documents defining:
 - a) the Applicant's legal status;
 - b) the principal place of business;
3. **Survey Department** and its authorized representatives are hereby authorized to verify the statements, documents, and information submitted in connection with this application. This Letter of Application will also serve as authorization to any individual or authorized representative of any institution referred to in the supporting information, to provide such information deemed necessary and requested by yourselves to verify statements and information provided in this application, or with regard to the resources, experience, and competence of the Applicant.
4. **Survey Department** and its authorized representatives are authorized to contact any of the signatories to this letter for any further information.¹
5. All further communication concerning this Application should be addressed to the following person,

[Person]

[Company]

¹ Applications by joint ventures should provide on a separate sheet, relevant information for each party to the Application.



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[Address]

[Phone, Fax, Email]

6. We declare that, we have no conflict of interest in the proposed procurement proceedings and we have not been punished for an offense relating to the concerned profession or business and our Company/firm has not been declared ineligible.
7. We further confirm that, if any of our experts is engaged to prepare the TOR for any ensuing assignment resulting from our work product under this assignment, our firm, JV member or sub-consultant, and the expert(s) will be disqualified from short-listing and participation in the assignment.
8. The undersigned declares that the statements made and the information provided in the duly completed application are complete, true and correct in every detail.

Signed :

Name :

For and on behalf of (name of Applicant or partner of a joint venture):

A handwritten signature in black ink, appearing to be a stylized name, located at the bottom left of the page.

2. Applicant's Information Form

(In case of joint venture of two or more firms to be filled separately for each constituent member)

1. Name of Firm/Company:
2. Type of Constitution (*Partnership/ Pvt. Ltd/Public Ltd/ Public Sector/ NGO*)
3. Date of Registration / Commencement of Business (*Please specify*):
4. Country of Registration:
5. Registered Office/Place of Business:
6. Telephone No; Fax No; E-Mail Address
7. Name of Authorized Contact Person / Designation/ Address/Telephone:
8. Name of Authorized Local Agent /Address/Telephone:
9. Consultant's Organization:
10. Total number of staff:
11. Number of regular professional staff:

(Provide Company Profile with description of the background and organization of the Consultant and, if applicable, for each joint venture partner for this assignment.)



3. Experience

3(A). General Work Experience

(Details of assignments undertaken. Each consultant or member of a JV must fill in this form.)

S. N.	Name of assignment	Location	Value of Contract	Year Completed	Client	Description of work carried out
1.						
2.						
3.						
4.						
5.						
6.						
7.						



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3(B). Specific Experience

Details of similar assignments undertaken in the previous seven years

(In case of joint venture of two or more firms to be filled separately for each constituent member)

Assignment name:	Approx. value of the contract (in current NRs; US\$ or Euro) ² :
Country: Location within country:	Duration of assignment (months):
Name of Client:	Total No. of person-months of the assignment:
Address:	Approx. value of the services provided by your firm under the contract (in current NRs; US\$ or Euro):
Start date (month/year): Completion date (month/year):	No. of professional person-months provided by the joint venture partners or the Sub-Consultants:
Name of joint venture partner or sub-Consultants, if any:	Narrative description of Project:
Description of actual services provided in the assignment: Note: Provide highlight on similar services provided by the consultant as required by the EOI assignment.	

Firm's Name: _____

² Consultant should state value in the currency as mentioned in the contract



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3(C). Geographic Experience

Experience of working in similar geographic region or country

(In case of joint venture of two or more firms to be filled separately for each constituent member)

No	<i>Name of the Project</i>	<i>Location (Country/ Region)</i>	<i>Execution Year and Duration</i>
1.			
2.			
3.			
4.			
5.			
6.			
7.			



4. Capacity

4(A). Financial Capacity

(In case of joint venture of two or more firms to be filled separately for each constituent member)

Annual Turnover	
Year	Amount Currency

- Average Annual Turnover of Best of 3 Fiscal Year Of Last 7 Fiscal Years

--

(Note: Supporting documents for Average Turnover should be submitted for the above.)



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4(B). Infrastructure/equipment related to the proposed assignment³

No	Infrastructure/equipment Required	Requirements Description
1.	LiDAR Aircraft (1)	
2.	LiDAR Post Processing Facility	
3.	GPS Equipment (GPS and DGPS) (at least 8 equipment's)	

³ Delete this table if infrastructure/equipment for the proposed assignment is not required.



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5. Key Experts *(Include details of Key Experts only)*

(In case of joint venture of two or more firms to be filled separately for each constituent member)

SN	Name	Position	Highest Qualification	Work Experience (in year)	Specific Work Experience (in year)	Nationality
1						
2						
3						
4						
5						

(Please insert more rows as necessary)



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1. CV of Key Experts to be involved in the proposed service:

(Note: CV of each Key Experts to be involved in the proposed service should be submitted)

Proposed Position: _____

Name of Consultant: _____

Name of Staff: _____

Profession: _____

Date of Birth: _____

Years with Consultant/Entity: _____ Nationality: _____

Membership in Professional Societies: _____

Detailed Tasks Assigned: _____

Key Qualifications: _____

[Give an outline of staff member's experience and training most pertinent to tasks on assignment. Describe degree of responsibility held by staff member on relevant previous assignments and give dates and locations. Use about half a page.]

Education: _____

[Summarize college/university and other specialized education of staff member, giving names of schools, dates attended, and degrees obtained. Use about one quarter of a page.]

Employment Record: _____

[Starting with present position, list in reverse order every employment held. List all positions held by staff member since graduation, giving dates, names of employing organizations, titles of positions held, and locations of assignments. For experience in last ten years, also give types of activities performed and client references, where appropriate. Use about two pages.]

Languages: _____

[For each language indicate proficiency: excellent, good, fair, or poor in speaking, reading, and writing.]

Certification:

I, the undersigned, certify that to the best of my knowledge and belief, these data correctly describe me, my qualifications, and my experience. _____

Date: _____

[Signature of staff member and authorized representative of the consultant]

Day/Month/Year

Full name of staff member: _____

